

The Inequalities in Educational Outcomes Across Wards in the Northwestern Regions of Merseyside and Greater Manchester

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Abstract (Suji)

This report visualizes insights into socioeconomic geography by using multi-dimensional measures to understand the inequalities in educational outcomes between districts in Northwestern UK metropolitan areas (Merseyside, Greater Manchester) at the local ward level. It reveals that diverse socioeconomic backgrounds impact educational outcomes within homogeneous areas, which may necessitate further policy actions for balanced development and growth across the UK regions.

1. INTRODUCTION (Meghan)

The topic of educational disparities among populations is not a new concept being analyzed, and that is evidenced by the plethora of academic articles exploring variables that influence and are influenced by academic achievements. In this assignment, we were tasked with researching and analyzing datasets and variables that could affect educational outcomes in the 2021 UK Census data. Topological data analysis has proved useful when it comes to modeling and analyzing large and complex datasets. In contrast to the small and simple datasets comprised of few observations from past scientists that could be analyzed by hand, researchers today are consistently dealing with large and complex datasets that require more powerful and nuanced techniques to properly process the data (Carlsson, 2020). Because deprivation is based in the combination of different factors affecting a certain group, TDA's ability to find connections and provide subgroups of data that traditional clustering methods may fail to recognize (Godwin et al., 2019). In other words, TDA is important when analyzing deprivation because it can find otherwise hidden connections that can help influence decisions made in regard to the datasets.

Instead of looking at the whole of the UK for influences of educational outcomes, we are only looking at the areas of Greater Manchester and Merseyside. Together, the area we are looking at is comprised of 297 different wards that make up 14 local authorities (LSOAs). This project requires us to analyze these large areas at the ward level, which gives more

targeted data, allowing us to get the most precise results possible.

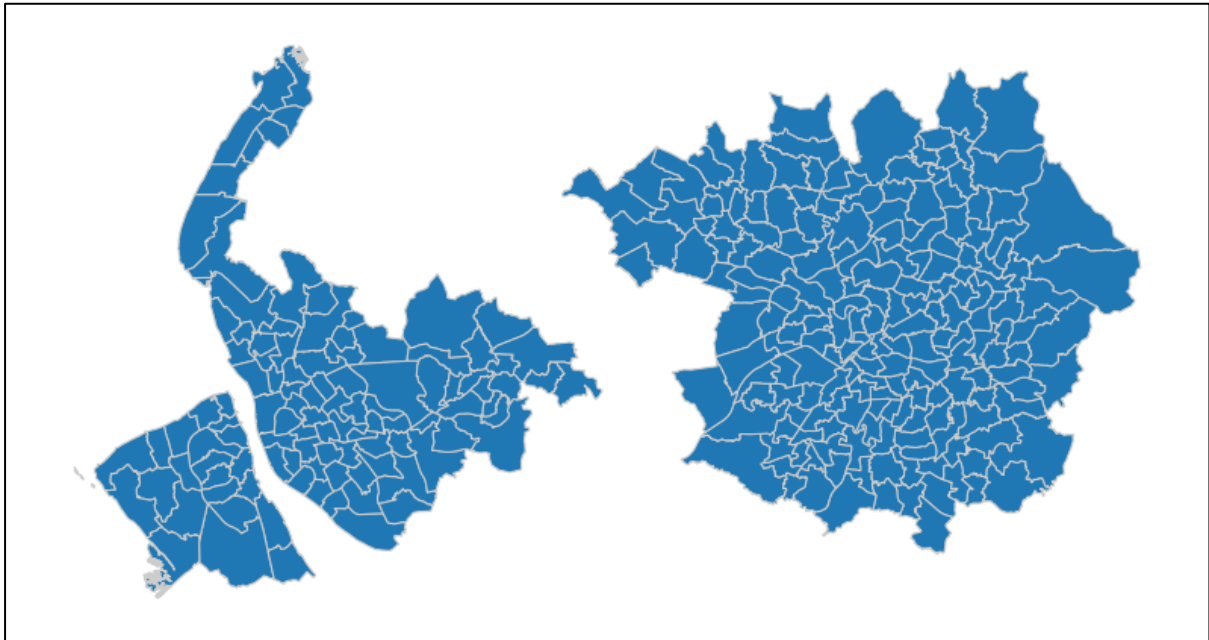


Figure 1. Illustration of Greater Manchester and Merseyside Wards

To get an initial grasp of the spatial data, Topological Data Analysis Ball Mapper (TDABM) algorithm developed by Dłotko (2019a) is utilized to showcase the distribution of qualifications across the areas of Greater Manchester and Merseyside, as shown in Figure 2.

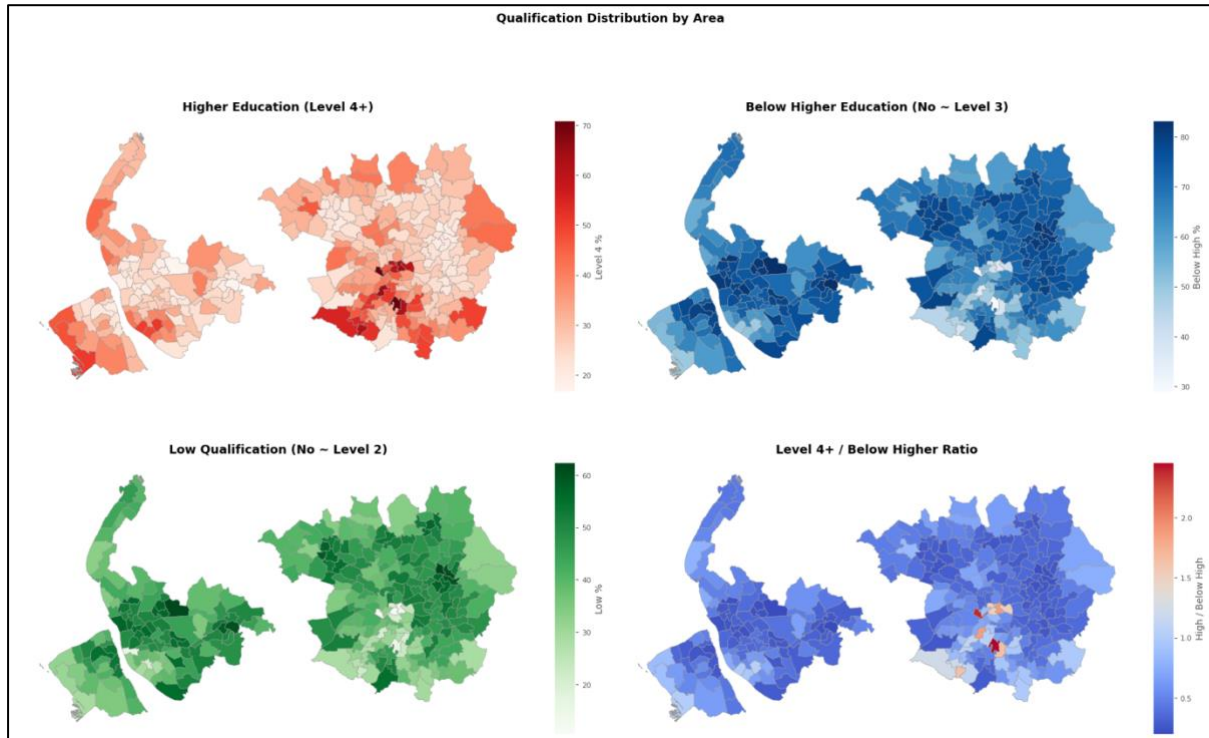


Figure 2. Distribution of Qualifications

Observed trends in regional qualification distributions while not establishing causality, facilitated initial data inferences. The Ball Mapper algorithm, by visualizing 28-dimensional space, aided in discerning regional data homogeneity or heterogeneity.

In this study, educational inequality is assessed using two primary outcome variables: the attainment of university-level qualifications or higher, and the completion of compulsory secondary education. This approach reflects the post-compulsory educational paradigm within the UK's developed socioeconomic context, where secondary education is mandated (UK Government, 2012). Although other metrics, such as the Gini coefficients of education (Thomas et al., 2001), were considered to capture multifaceted aspects of educational inequality, their primary applicability to developing countries rendered them unsuitable for this analysis within the UK context.

During data preprocessing, we consolidated all analytical factors into a unified dataset, enabling efficient variable extraction. Throughout integration, precise variable categorization was maintained, particularly between council and social housing classifications. Social housing was included due to its affordability compared to market rates and spatial

concentration; characteristics deemed analytically significant.

2. LITERATURE REVIEW (Jiaqi and Meghan)

Previous studies summary

Much of the previous research surrounding this topic has explored the relationship between educational inequality and other socio-economic factors. The studies mostly focus on whether and how social class and economic background affect education. Wilkinson and Pickett (2009) argue that educational issues arise from the polarization of the distribution of class within society. Reay (2017) adds that lower class students are facing relatively larger significant barriers to academic success. Studies by Evans (2003), Chen et al. (2010), and Blair et al. (2011) prove that experiencing poverty has negative impacts on children's early brain development and cognitive abilities, which in turn affect their education outcome and academic success. Some researchers are focusing on other socio-economic factors. Research from Gillborn et al. (2016) highlights persistent educational achievement gaps between different racial and ethnic groups. There are also researchers discussing the impact of the education system (Tahir, 2022), education infrastructure (Cottrell-Boyce, 2024) and educational market (Gibbons et al. (2008), over educational inequality. In the Suhreke et al. (2011) analysis of how health behaviors affect educational outcomes, they found that in the high-income countries observed, there was a correlation between a person's health status and educational achievements. The concept that household conditions could positively influence educational outcomes was explored by Lien et al. (2008), and although they did not find strong evidence that housing status affects educational outcomes, they observed that adolescents in certain housing situations might experience challenges that influence their academic performance.

Another related focus of previous studies is geographic and regional disparities in education inequality. Farquharson et al. (2022) addresses that the academic performance of students that live in the southern part of the UK is better than students who are from the northern parts or more remote areas. Playford et al. (2023) concludes that students from rural

and coastal areas face greater challenges in accessing educational resources, after performing a multivariate regression analysis. Machin et al. (2013) details that since each country within the UK has its own school system, educational requirements and outcomes can vary spatially as well. One of the popular explanations for the spatial distribution of educational outcomes is the idea that urban areas drive educational involvement for the generations living there. Van Maarseveen (2021) observed a collection of students in the Netherlands and how they interacted with the Dutch education system, specifically regarding their school classification choice after primary and secondary schools. He observed that “children who grew up in urban regions consistently achieve higher levels of human capital compared to children in rural areas”. He also observed that family characteristics, such as income or education, do not have a significant effect on the child’s educational pathway. If living in an urban environment has a positive correlation with educational achievement, but is possibly not due to their familial situation, it begs the question of what variables are influencing higher levels of educational outcomes.

Controversial issues and contesting claims in education inequality

There is an ongoing debate over whether there is a causal relationship between socio-economic factors and education inequality. Some are using methods such as natural experiments, instrumental variables and regression models to identify causal links. For example, Blanden et al.'s (2022) study examines the impact of household income on children's achievement, and Cheng and Williams' (2020) study examines how educational investment affects the outcomes. While other scholars remain skeptical. Hobbs (2016) emphasizes the challenges of drawing causal inferences between socio-economic factors and educational inequality.

Possible research gaps

While the ongoing debate remains, methods like topological data analysis, which we chose to apply, do not focus on causal inference but more on pattern and structure. Moreover, previous studies mostly apply theoretical analysis, with fewer studies applying other methods like natural experiment and instrumental variables. Topological data analysis may provide

new perspectives and methods for the study of educational inequality, and it may offer new insights to study educational inequality in a more data-driven way. Additionally, our analysis may enrich the existing literature by providing a more comprehensive view. Although there is a large amount of research on educational outcomes in the UK, the current literature mainly focuses on the relationship between educational inequality, social mobility, and economic backgrounds (income, educational investment, etc.). Our analysis uses ward-level data from Merseyside and Greater Manchester, trying to expand the focus by using more socio-economic factors. The variables derived from health, disability, deprivation, tenure, economic activity, ethnic group, socio-economic classification, housing, car ownership, age, and number of schools.

3. METHODOLOGY (Jahanvi)

Topological data analysis is a new and growing field of data analysis which is backed by mathematical techniques and computational tools (Chazal & Michel, 2021). It explores topology and geometry in a way that can be used to infer robust information (qualitative or quantitative) about the structure of complex data. It is extremely versatile and has been used in areas of transportation (Li et al., 2019), chemistry (Smith et al., 2021), material science (Pike et al., 2020), genomics (Carrière and Rabadán, 2020), and medicine (Dindin et al., 2020) to name a few. It plots data as a collection of points (resembling a scatterplot when two-dimensional data is used). However, TDA falls behind when the dimensionality of the data increases. The topological data analysis ball mapper (TDABM) provides a solution to this problem by creating an abstract representation of the collection of data points while keeping the preexisting connections in the dataset (Rudkin et al., 2023).

TDABM starts by choosing a random point from the dataset and creating a ball of radius ϵ around it. If the dataset only consists of two dimensions, the ball will be visualized as a circle. However, TDABM holds the capacity to work with any n number of dimensions (Rudkin et al., 2023). The first point with radius ϵ becomes a landmark and all other points that come under the ball's radius are included within the ball. This process continues as the algorithm chooses another uncovered point randomly and creates a ball around it, incorporating all the

points that fall within it. It only ends when all the points in the dataset are covered by a ball. It is likely that some balls may intersect with each other. This intersection is what gives the balls their relative position. If there exists a point in the intersection of two balls, an edge is drawn. The size of the balls is calculated by the number of data points it covers, with larger balls meaning more points within radius ϵ . The colour of the ball refers to summary statistics of the dataset which can be chosen according to the research question. One of the most important decisions to make with TDABM is the choice of radius. There is no “perfect” value of ϵ , however, it is recommended to use multiple radii to choose the best one for your data (Rudkin et al., 2023). Throughout this report, different values of ϵ are used to keep the number of balls between 20-30 enabling more in-depth analysis of the different wards. Based on the visualizations, certain balls were chosen for closer analysis. Using TDABM in the current study helps us to analyze the trends within different wards in Merseyside and Manchester.

4. DATA (Jahanvi)

The datasets used in this report come from a large variety of sources. Our main area of focus is the highest qualification variable, represented as a percentage, as seen in the 2021 UK census and aggregated at the ward level (NOMIS, 2021). Only wards within Merseyside and Greater Manchester were selected. Qualifications are distributed into four levels starting from entry level qualifications up to work-related qualifications. The datasets that we decided to explore with qualifications can be broadly distributed into the following categories (See Table 1).

Category	Dataset
Highest level of qualification	data_qualification (NOMIS, 2021)
Life expectancy	data_generalhealth (NOMIS, 2021)
	data_disability (NOMIS, 2021)
Index of multiple deprivation	data_deprivation (NOMIS, 2021)
	data_tenure (NOMIS, 2021)
	data_economicactivitystatus (NOMIS, 2021)
Ethnicity	data_ethnicgroup (NOMIS, 2021)
Social class	data_ns_sec (NOMIS, 2021)
	data_socialstatus_occupation (NOMIS, 2021)
Open space	GB_Greenspace (OS Data Hub, n.d.)
Population density	Population density (NOMIS, 2021)
Car access	data_car (NOMIS, 2021)
Overcrowding in households	data_nobedrooms (NOMIS, 2021)
	data_norooms (NOMIS, 2021)
	data_occupancybedrooms (NOMIS, 2021)
	data_occupancyrooms (NOMIS, 2021)
Age profile	data_age (NOMIS, 2021)
School Information	School_Locations_UK_2025 (NOMIS, 2021)

Table 1: Variables and Datasets

Over 15 datasets were referred, and a dataset with 297 rows and 317 columns was formed. This dataset was then edited and refined by concatenating columns, deleting irrelevant and duplicate information, and normalizing variables, giving equal importance to all (See APPENDIX). Most of the variables in our dataset were already present in a range from 0-100%. Therefore, only some variables were scaled (for example, number of schools and population density). The 28 columns used for our analysis are mentioned in Table 2. It is important to note that all 28 variables were only used for our robustness tests. The variables used for ball mapper only include 10 out of these 28, namely, Ndisabledp, Deprived24p, socrentp, nonemployedp, ethnotherp, manage_nssec, nocarp, midhousingp, oldp,

numberofschools (See APPENDIX for reasoning).

Group	New Variable	Description
Qualification	high	Percentage of population with Level 4 or above qualification
	below_high	Percentage of population with below higher education qualifications (including Level 1, Level 2, Apprenticeship, Level 3, and Other qualifications)
	low	Percentage of population with low education qualifications (including Level 1 and Level 2 qualifications)
	ratio	Ratio of population with higher qualifications (Level 4 or above qualifications) to below higher qualifications (Level 1, Level 2, Apprenticeship, Level 3 and other qualifications)
Health	goodhealthp	Percentage of population reporting very good or good health
Disability	ndisabledp	The sum of the percentages of people with disabilities (including those under the Equality Act, limited a lot, and limited a little.)
Deprivation	Deprived0p	Percentage of households not deprived in any dimension
	Deprived1p	Percentage of households deprived in one dimension
	Deprived24p	Percentage of households deprived in two or more dimensions
Tenure	tenureownp	Percentage of population who own homes outright
	socrentp	Percentage of population in social rented homes
	privrentp	Percentage of population in private rented homes
Economic Activity	employeep	The percentage of the population that is employed (including full-time or part-time)
	selfemployeep	Percentage of self-employed individuals

		(including with and without employees)
	nonemployedp	Percentage of non-employed individuals (including unemployed)
Ethnic Group	ethnwhitep	Percentage of population identified as White
	ethnotherp	Percentage of population identified as other ethnic groups (including Asian, Black, Mixed, or Other ethnic groups)
Socio-economic classification	nssec1	Percentage of population in higher managerial/professional occupations
	nssec2	Percentage of population in lower managerial/intermediate occupations
	nssec3	Percentage of population as self-employed or small employers
	nssec4	Percentage of population in semi-routine or routine occupations
	nssec5	Percentage of population unemployed or full-time students
Housing	lowhousingp	Percentage of population in low housing conditions (1-2 rooms)
	midhousingp	Percentage of population in mid-range housing conditions (3-5 rooms)
	highhousingp	Percentage of population in high housing conditions (6+ rooms)
Car Ownership	nocarp	Percentage of population with no car ownership
	1carp	Percentage of population with one car
	23carp	Percentage of population with two or more cars
Age	youthp	Percentage of population aged 4-15 years
	adultp	Percentage of population aged 16-64 years
	oldp	Percentage of population aged 65 years and over
Schools	numberofschools	Number of schools in the area

Table 2: Variables Used in the Analysis

The descriptive statistics for these are mentioned in Table 3.

Variable	Count	Mean	SD	Min	Q1 (25%)	Q2 (50%)	Q3 (75%)	Max
goodhealthp	297	80.28	3.69	70.7	78	80.2	82.4	93.2
disabledp	297	80.40	3.60	70.5	78.5	80.6	82.7	89.8
Deprived0p	297	45.21	8.81	23.5	38.6	44.9	51	70.3
Deprived1p	297	33.12	2.19	24.1	32	33.4	34.5	37.9
Deprived24p	297	21.65	7.47	5	16.1	21	27.1	41.5
tenureownp	297	59.18	18.02	14.5	47	61.7	73.5	90.7
socrentp	297	20.09	12.18	1.6	10	17.9	27.4	55.8
privrentp	297	19.95	11.06	6.2	12.9	16.8	23.4	78.4
employeeep	297	45.82	5.86	22.4	43.3	46.5	49.4	66.6
selfemployeeep	297	7.68	1.56	3	6.6	7.5	8.5	14
nonemployedp	297	44.07	5.53	19.2	40.8	43.9	47.2	58.1
ethnwhitep	297	82.24	17.90	16.5	77.1	89.1	94.8	98.2
ethnotherp	297	17.76	17.89	1.9	5.3	10.9	22.9	83.4
nssec1	297	11.06	5.92	2.6	6.7	9.6	13.9	33.1
nssec2	297	31.31	6.63	12.6	26.8	32.3	36.5	42.9
nssec3	297	13.97	2.02	4.1	13.2	14.4	15.2	17.2
nssec4	297	24.61	7.03	5.3	19.8	25.4	30.1	41.1
nssec5	297	19.04	9.21	8.9	13.2	15.9	21.8	68.1
popdensity	297	25.77	16.21	0	13.83	24.77	35.47	100
nocarp	297	27.78	11.85	8.3	18.8	25.8	35.3	68.1
1carp	297	42.53	3.09	27.6	41	42.9	44.5	50.2
23carp	297	29.68	10.95	4.3	21.6	29.9	37.7	57.5
lowhousingp	297	33.51	14.63	7	23.2	31.4	41.5	94.8
midhousingp	297	62.35	13.82	5	54.4	63.4	72.2	90.7
highhousingp	297	4.12	3.96	0	1.6	2.9	5.3	29.1
youthp	297	19.35	4.23	2.32	16.99	19.07	21.49	30.55
oldp	297	17.40	6.13	1.80	13.45	17.40	21.19	37.04
numberofschools	297	7.52	4.94	0	4	7	10	27

Table 3: Summary Statistics

A correlation matrix was also constructed to understand the relationship between variables and remove any issues of multicollinearity.

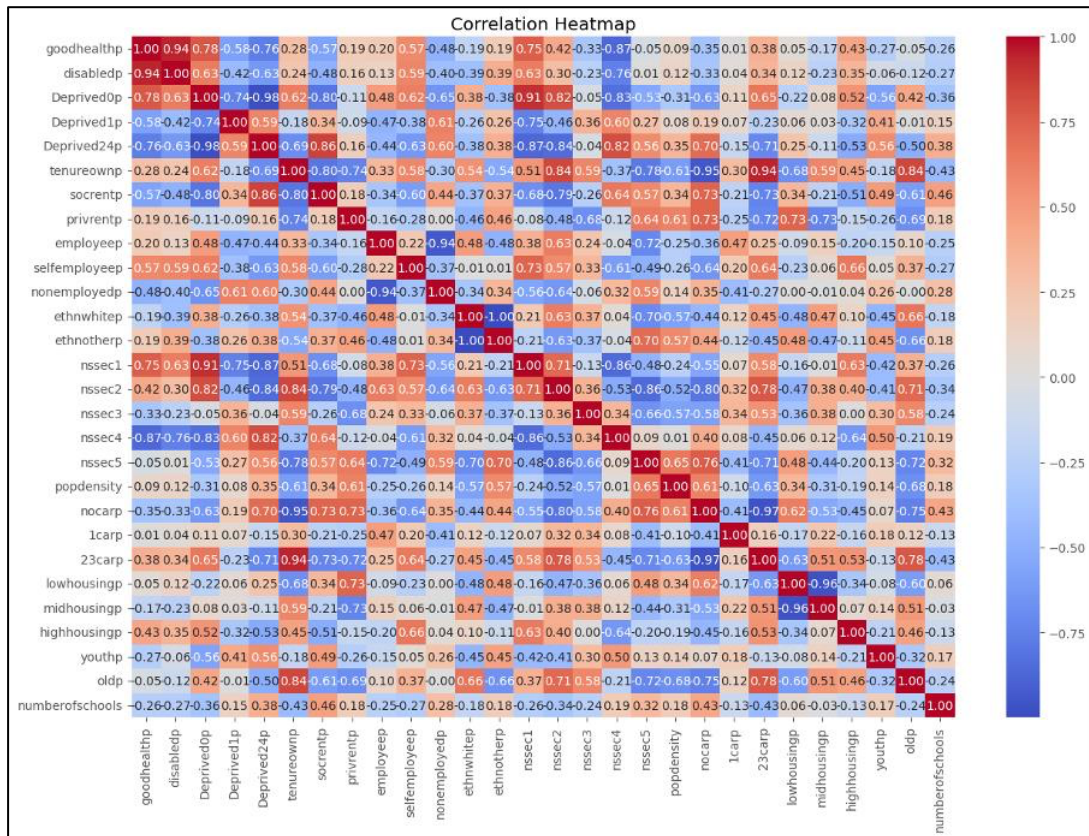


Figure 3: Correlation Heatmap of all Variables

The correlation matrix for the 10 variables used for ball mapper is also drawn.

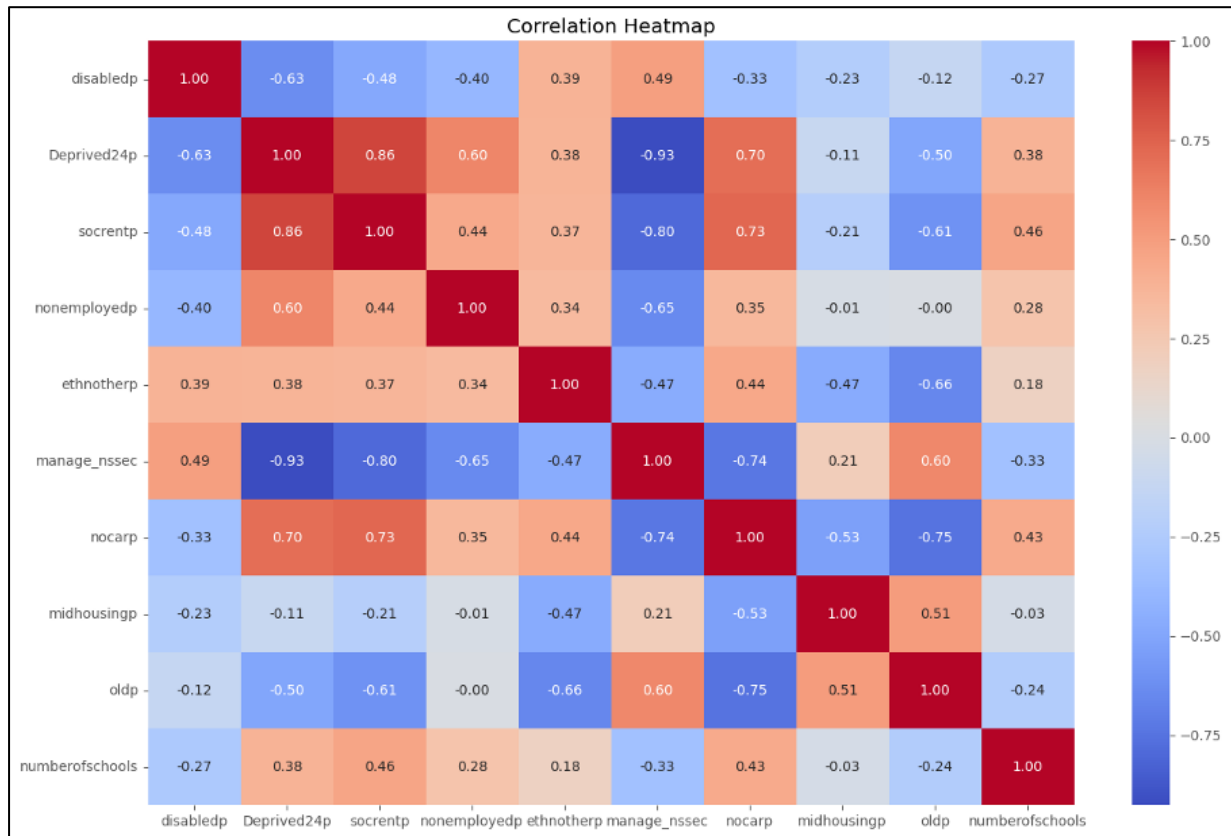


Figure 4: Correlation Heatmap with 10 variables

Some preliminary graphs were made to understand the distribution of data at the local area level. The average population density of each area is observed through the following plot with Manchester and Liverpool having the highest population and St Helens having the lowest.

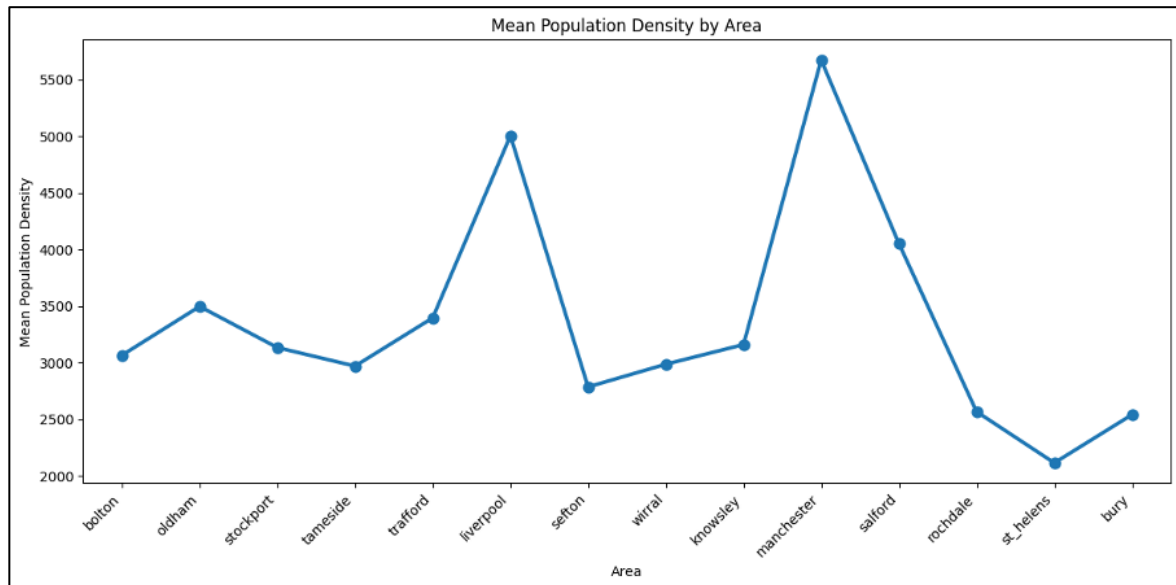


Figure 5: Point plot for Mean Population Density by Area

The distribution of qualifications per area clearly shows us the lower number of high qualifications (denoted by the blue bar). A similar trend can be seen in the education and training statistics report (GOV.UK, 2024) which shows a sharp decline in the number of students in school vs higher education.

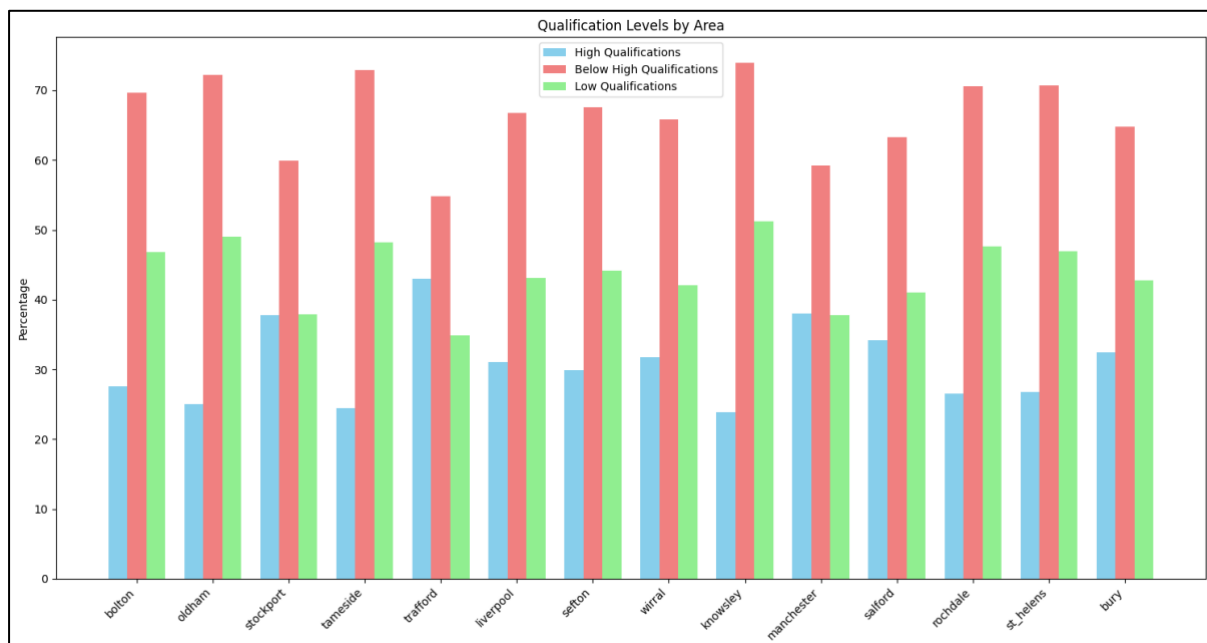


Figure 6: Bar Chart for Qualification levels by Area

Zooming into these areas, the following trends were seen at the ward level.

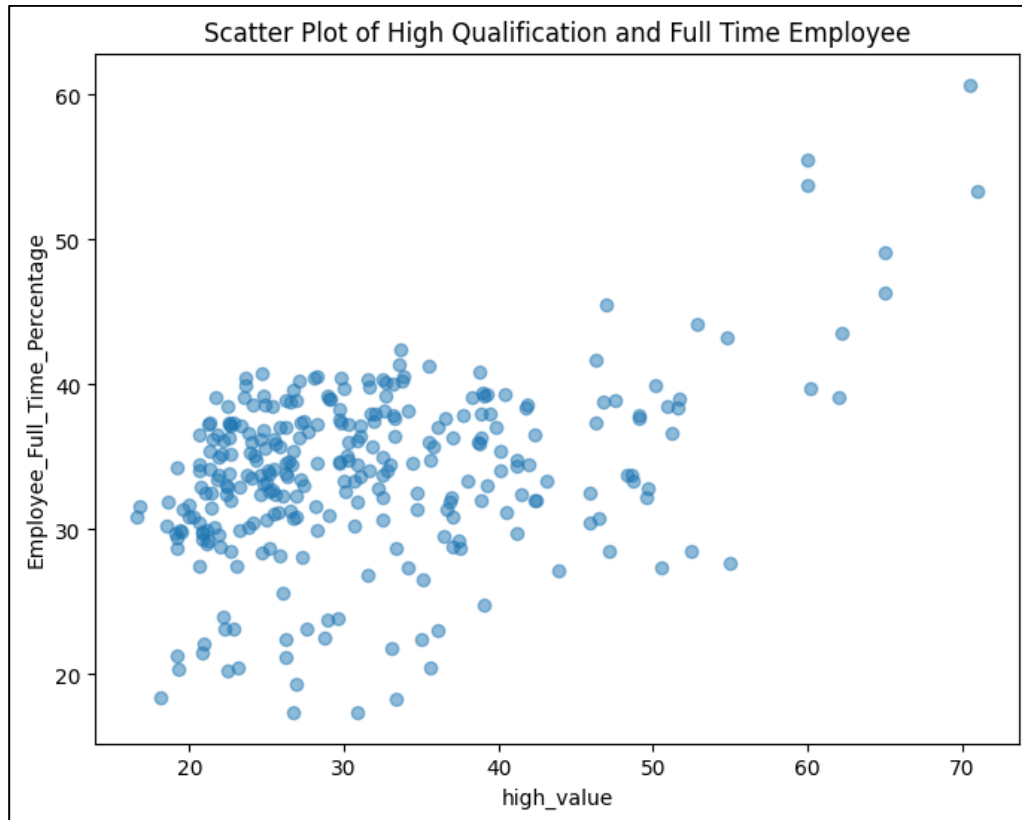


Figure 7: Scatter plot for Full Time Employee and High Qualification

Taking the full-time employee variable and plotting it against high qualification, we can see a positive relationship between the two. We can therefore infer that the percentage of people employed in full-time roles is related with high qualifications. Another scatterplot between car access and high qualification rate shows us owning a car does not relate to having a higher education. As the number of households with one car increases, the high qualification rate remains low. We went in with the assumption that those who have access to a car will be more likely to have the means for higher education (for example, travelling to and from campus would be easier). However, this is not the case. Public transport has made it easier for individuals to travel without direct access to cars (Paulley et al., 2006).

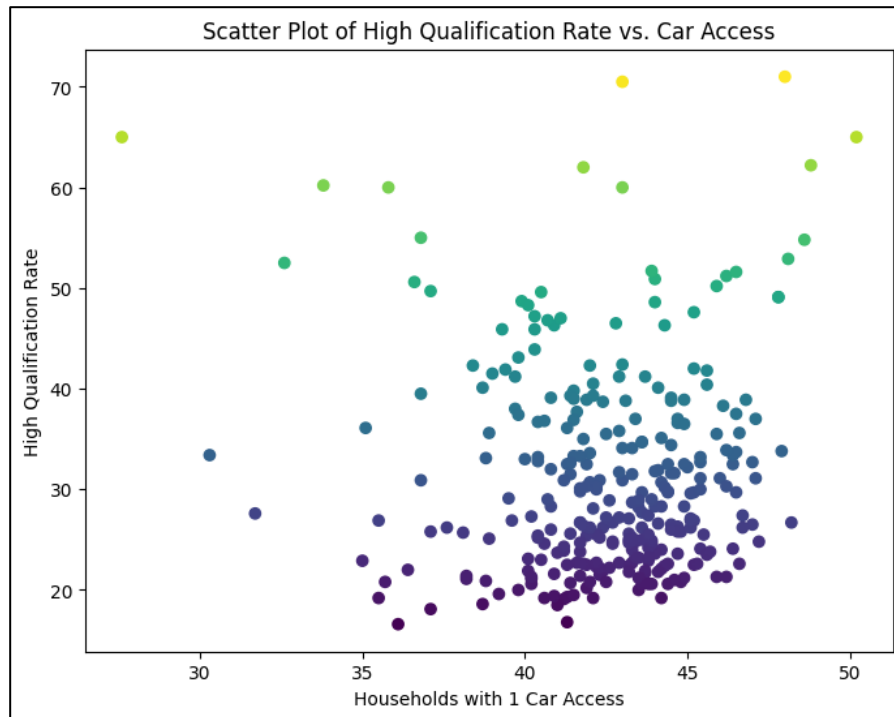


Figure 8: Scatterplot of Car access and High qualification Rate

We then move on to look at joint distributions. As we saw earlier, the mean population density was higher for Manchester and Liverpool. However, when we plot population density with high qualifications, we see that there is no linear relationship between the two.

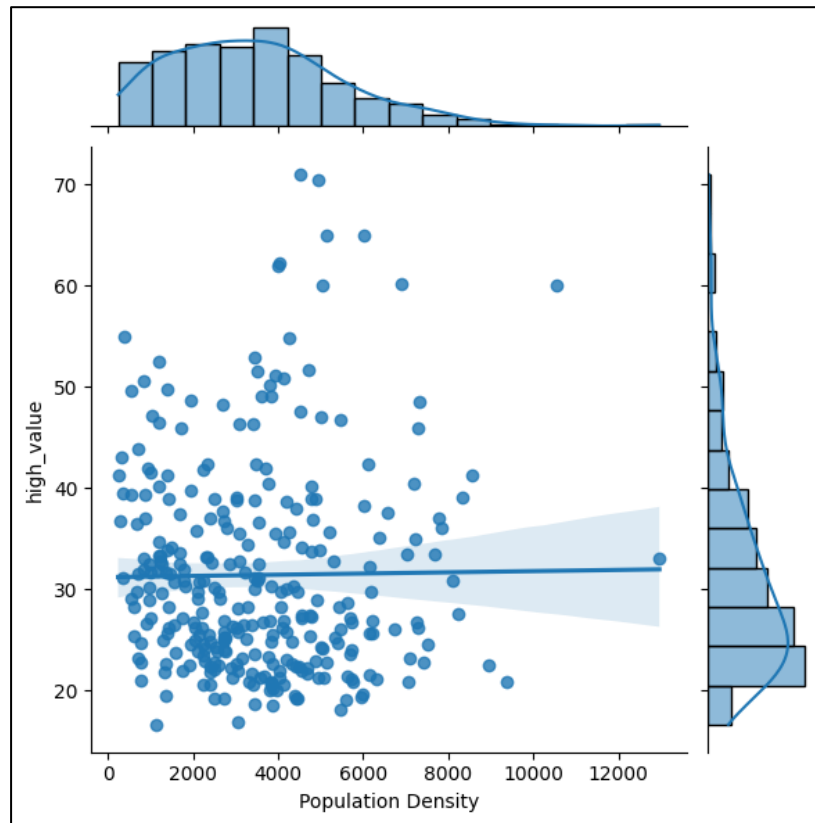


Figure 9: Joint Plot of Population Density with High Qualifications

Another joint plot with number of schools and high qualifications shows us an interesting downward trend. Since high qualifications include higher education (i.e., university level), it makes sense to assume that a higher number of schools in a ward does not lead to a higher number of Level 4 qualifications in the ward. This is explored more later in the report.

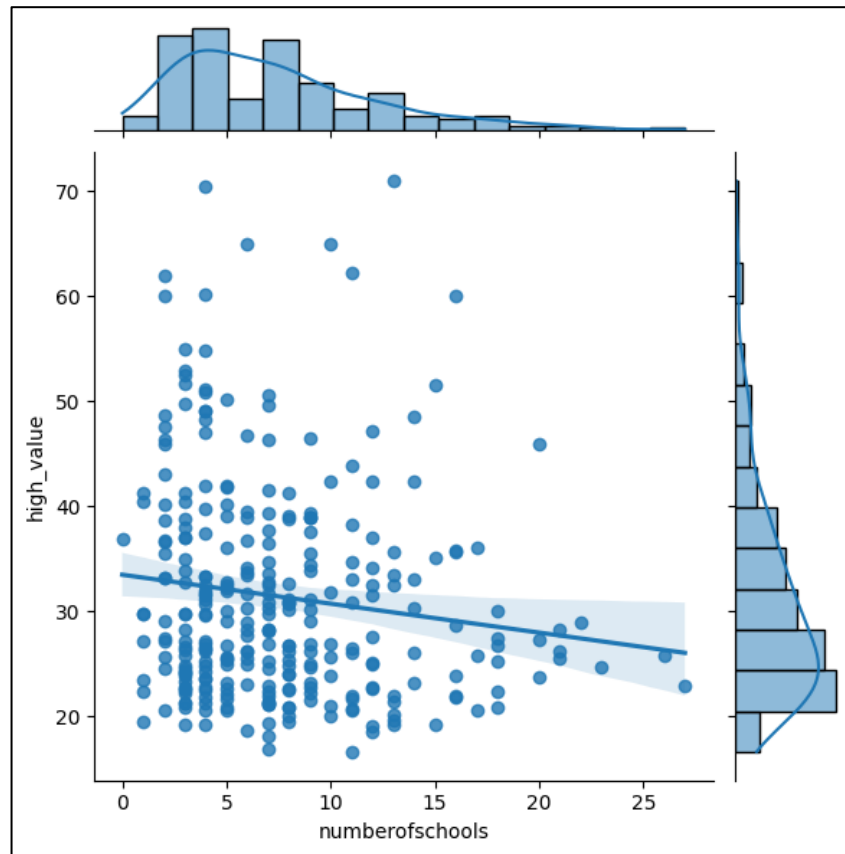


Figure 10: Joint Plot of Number of Schools and High Qualification

5. RESULTS: Ball Mapper Analysis (Suji)

The Ball Mapper graph ($\varepsilon = 25$) reveals the topological structure of a high-dimensional dataset with 28 ward-level features. About 30 balls—roughly 1/10 of the total wards—were used, reflecting the inverse relationship shown in Figure 11.

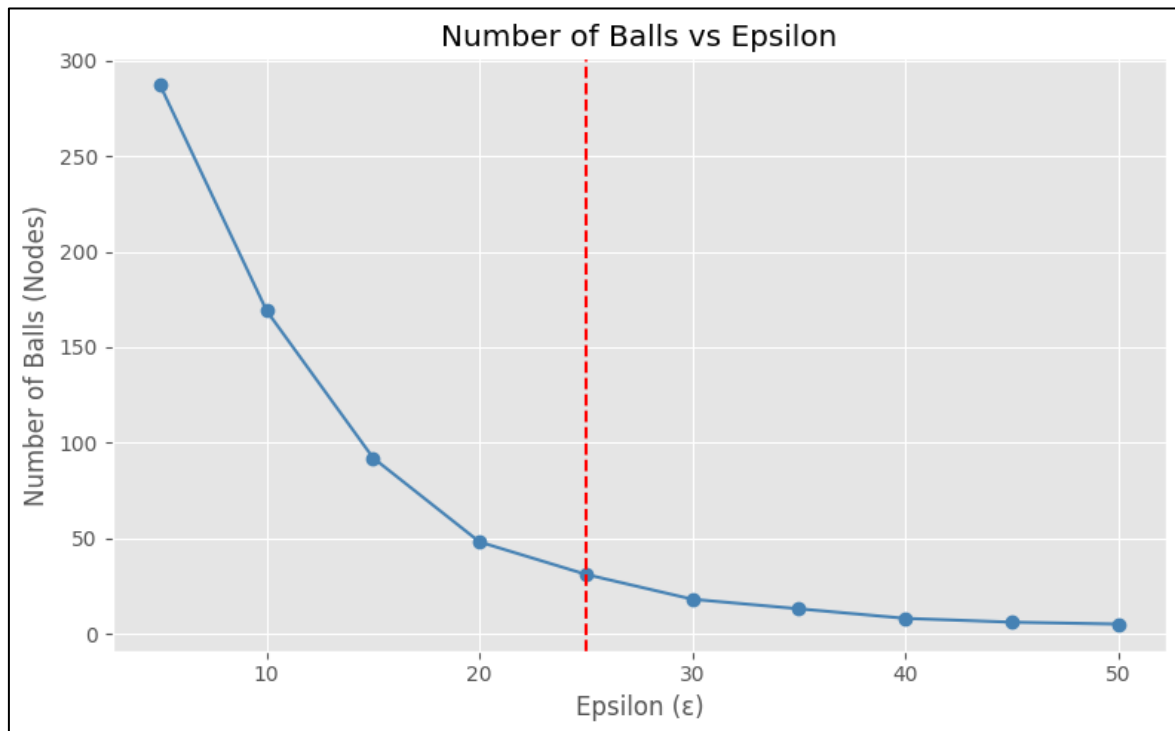


Figure 11. Relationship between the Number of Balls and Epsilon

The ball mapper topology shows nodal connections along a descending diagonal, with chromatic heterogeneity. Clusters with similar education levels are evident, while highly educated nodes remain topologically detached from the main structure.

The spatial distribution exhibits regionally heterogeneous low-education clusters, while centrally positioned balls function as topological bridges, connecting otherwise disparate educational profiles, suggesting the presence of shared characteristics beyond qualification metrics alone. A more comprehensive interpretative analysis of these structural patterns will follow.

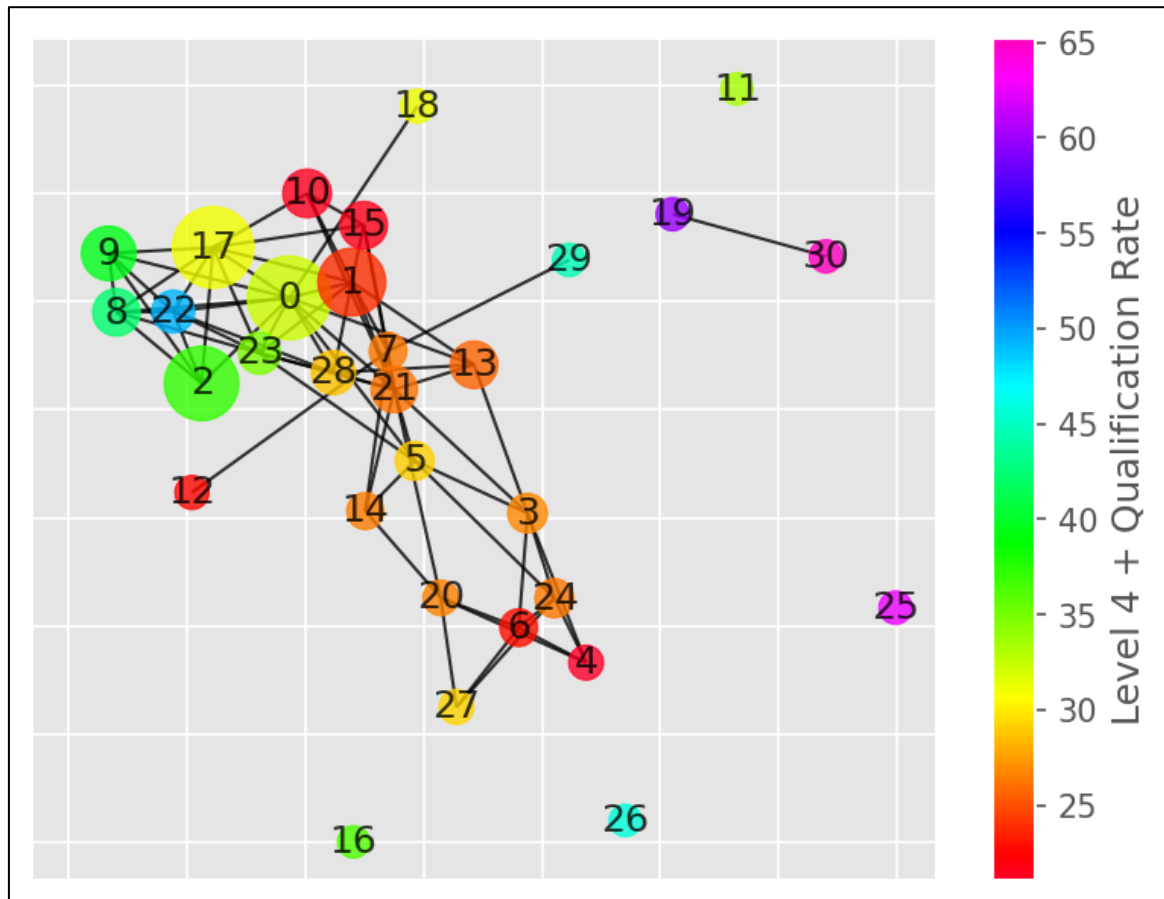


Figure 12. Ball Mapper Level 4 + Qualification rate

Table 4 shows the average high qualification rate and the number of wards in every balls. The fact that over 66% of all wards fall into balls 0, 1, and 2 out of 30 indicates characteristics including asymmetry or bias in the data distribution and represents the concentration of regional characteristics of each ward.

ball	high_valu e	numbe r	ball	high_valu e	Number	ball	high_valu e	numbe r
0	32.2	115	11	33.4	1	21	27.9	2
1	22.9	44	12	21.1	2	22	68.0	2
2	40.0	38	13	37.1	2	23	51.6	1
3	26.9	11	14	26.6	5	24	32.2	3
4	20.7	3	15	22.7	1	25	62.6	2

5	32.0	6	16	36.1	1	26	45.9	1
6	23.0	7	17	26.4	15	27	26.2	1
7	27.9	6	18	30.6	2	28	28.9	1
8	49.5	2	19	60.0	2	29	47.0	1
9	35.0	5	20	34.4	2	30	70.5	1
10	22.3	12						

Table 4. Average High Qualification Rate and Number of Datapoints in the Ball

The presence of these separated clusters indicates underlying heterogeneity within the data. To assess whether this reflects multivariate polarization across the variables included in the analysis, areas with high qualification rates ($\geq 50\%$) and low ($\leq 30\%$) were visualized separately (figure 0). The comparison revealed substantial disparities in deprivation, housing tenure, the proportion of non-employed individuals, occupational classification, and housing occupancy.

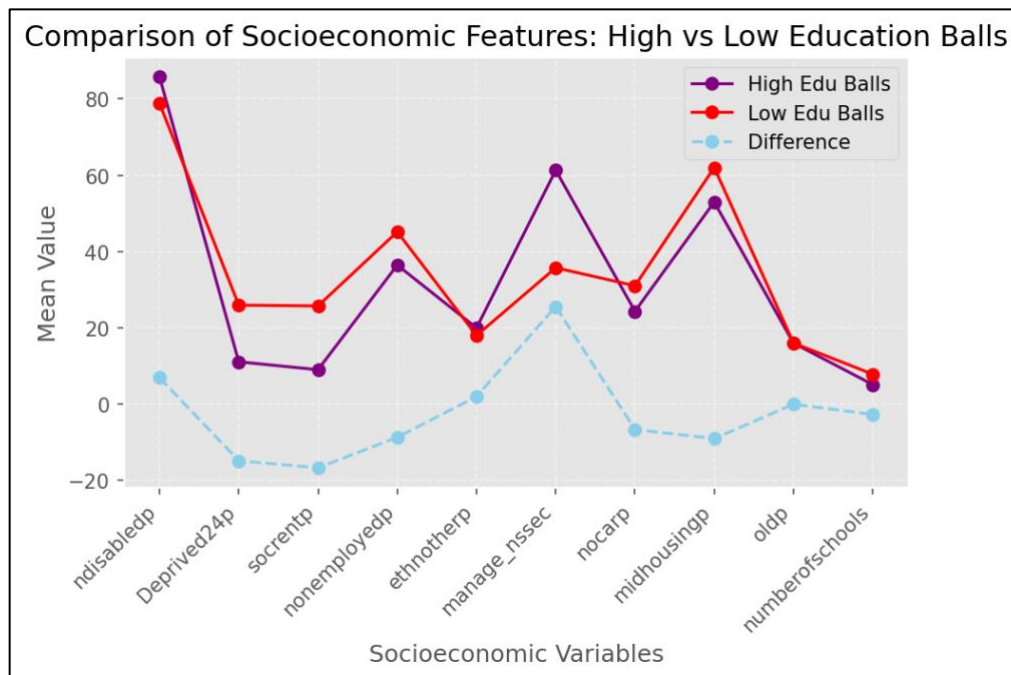


Figure 13. Feature Difference between the High Qualification and Low Qualification Balls

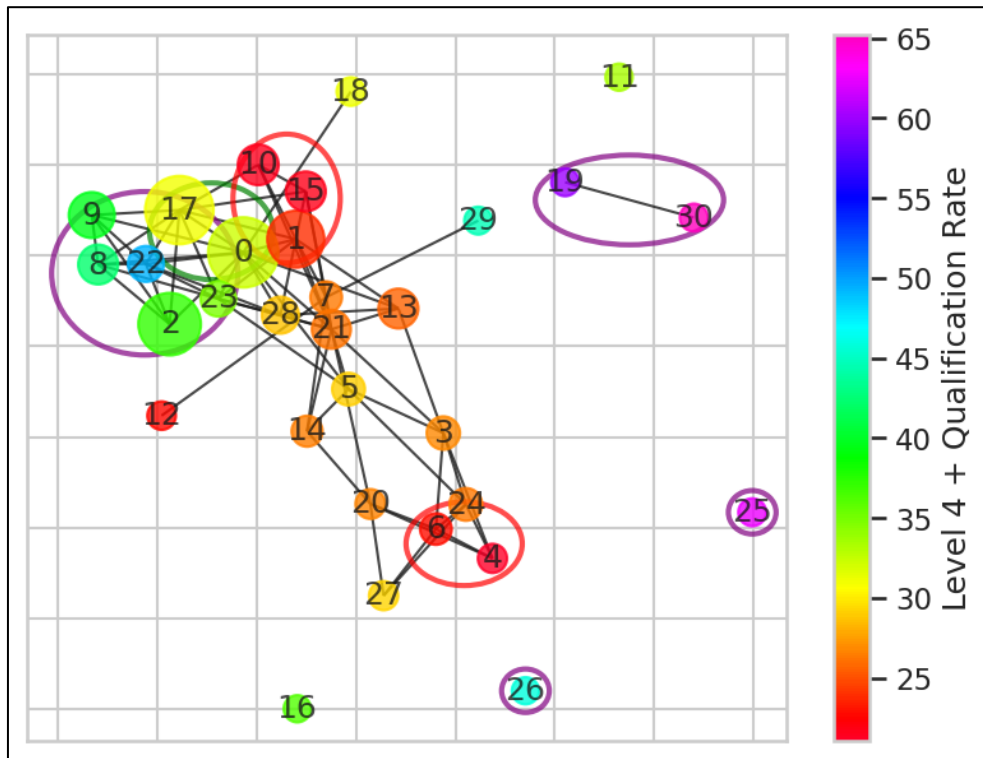


Figure 14. Level 4 + Qualification rate with boundaries

Purple Ellipse: High Qualification areas

Balls 2, 22, 23 form a densely connected cluster, high-qualification core (blue/green), with ball 2 dominant. These balls share higher managerial/professional occupation, a low disability rate, and a small number of social rent areas.

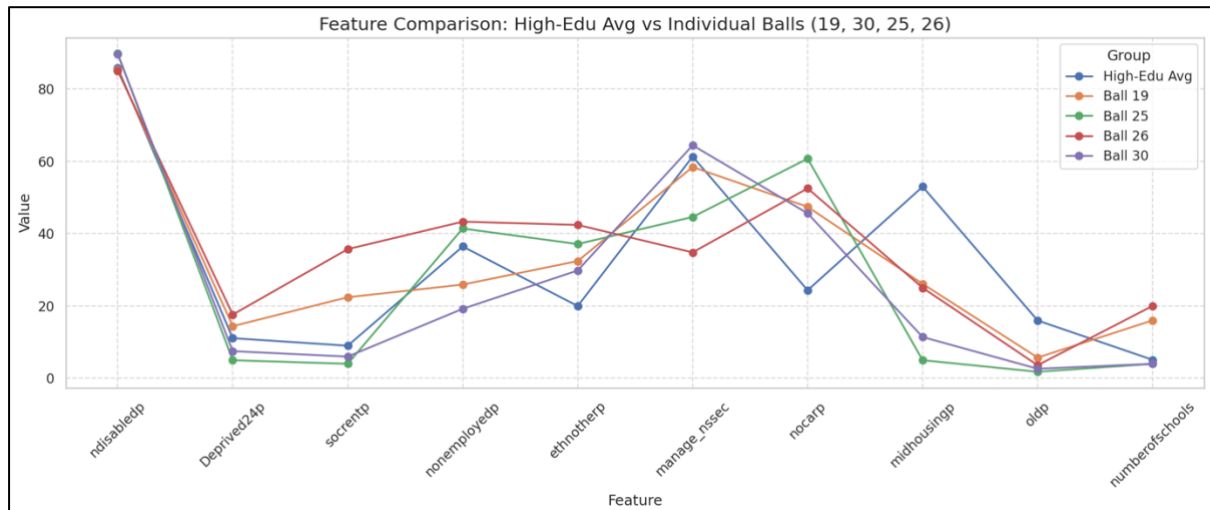


Figure 15: Ball 19, 30 (Connected) and Ball 25, 26 (Outliers) Comparison

For the concurrent evaluation of both heterogeneity and homogeneity characteristics within these nodes, a linear graphical representation was constructed; upon examination of this linear visualization (Figure 0), the quantitative values demonstrate a consistently uniform distributional pattern between ball 19 and 30, suggesting a niche similarity.

Ball 19 (Ancoats/Beswick, Blackfriars/Trinity) and 30 (Quays) average 60.6% managerial occupations. Ball 25 (Deansgate) features 64.4% no-car households and low prevalence of mid-sized housing (6.05%), typical of dense urban residential. Ball 26 (Hulme) has 35.7% social rentals (vs. 9.0% high-education average) and significantly fewer managerial professionals. These distinctive socioeconomic characteristics generate topological distance from the primary node, impeding network interconnectivity.

Conversely, balls 2, 8, 9, 22, 23, with less non-vehicle ownership and more mid-size housing, showed topological divergence, distancing them from primary clusters.

Green Ellipse: Bridge areas

Balls 17 and 0 function as topological bridges, linking high and low educational attainment clusters within the upper-left quadrant. This tells us that wards with both high and low educational attainment share several overlapping characteristics. Ball 17 encompasses 15 wards, while ball 0 comprises 115 wards, notably including Hyde Newton and Norwood. A salient distinction between these nodes lies in their ethnic composition; notably, Ball 17

exhibits a markedly low level of ethnic diversity, as evidenced by a non-White population proportion averaging 96.3%.

Red Ellipse: Low Qualification areas

To delineate geographical heterogeneity through the categorization of clusters—specifically, balls 10, 15, and 1, positioned at one extreme of the upper-left quadrant diagonal, forming one cluster, and balls 6 and 4, located at the opposing extreme, comprising another—it is observed that while these clusters share general attributes, notable distinctions emerge.

The cluster encompassing balls 10, 15, and 1, inclusive of wards such as Northwood, Norris Green exhibits a mid-sized housing proportion approximately double that of the other cluster, indicative of a suburban typology, and is characterized by a predominantly White demographic, constituting roughly 90%. Conversely, the 4 and 6 ball cluster, which includes Alexandra and St. Mary's wards, presents a higher prevalence of social rent housing, enhanced ethnic diversity (particularly in ball 6, where approximately 78.1% of the population is non-White), and a higher proportion of non-working individuals. However, the 10, 15, and 1 cluster demonstrates a higher degree of population ageing.

Upper Left Quadrant Diagonal

The main cluster is in the upper-left quadrant, with a branching structure extending diagonally down to the bottom right, making the connectivity and adjacency among nodes reflect similarity in multivariate profiles of the data points (Gibson et al., (2013)).

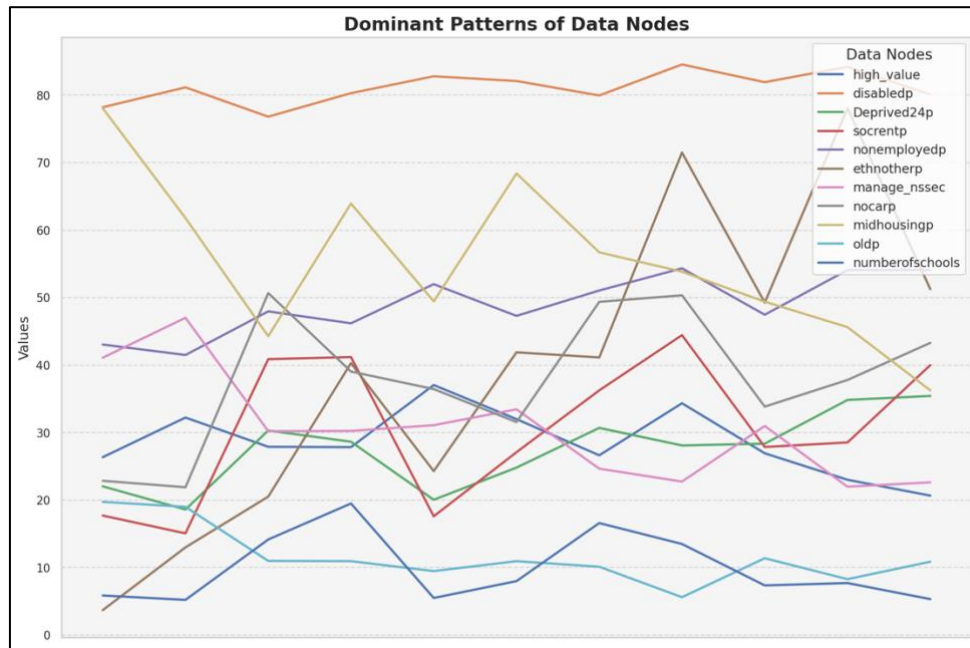


Figure 16: Dominant Patterns of Data Nodes

Examination of the figure reveals orange nodes extending diagonally rightward, depicting regions with continuous unidirectional variable variation. From balls 17 and 0 in the upper left quadrant with elevated educational attainment, a progressive diminution in educational levels is observable along the diagonal trajectory.

The Ball Mapper methodology corroborates that data points with comparable characteristics manifest continuous spatial distribution. (Höfling, F., & Franosch, T. (2013)) Through analysis of these characteristics utilizing 10 variables, as illustrated in Figure 0, it becomes evident that the directional pattern is predominantly influenced by demographic transitions, specifically: increased non-white populations, decreased mid-housing, higher non-car ownership, and more social rent.

6. ROBUSTNESS (Suji)

1) Hyperparameter tuning

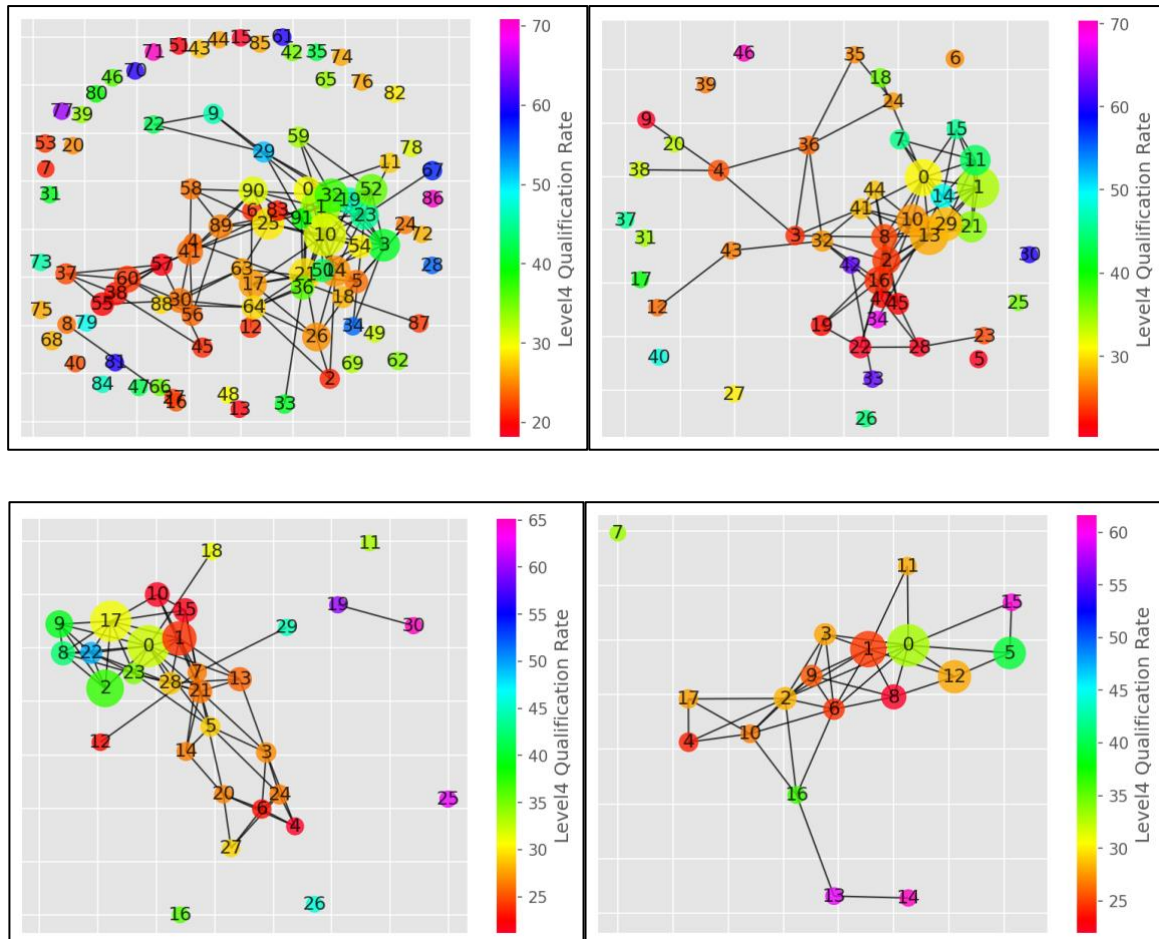
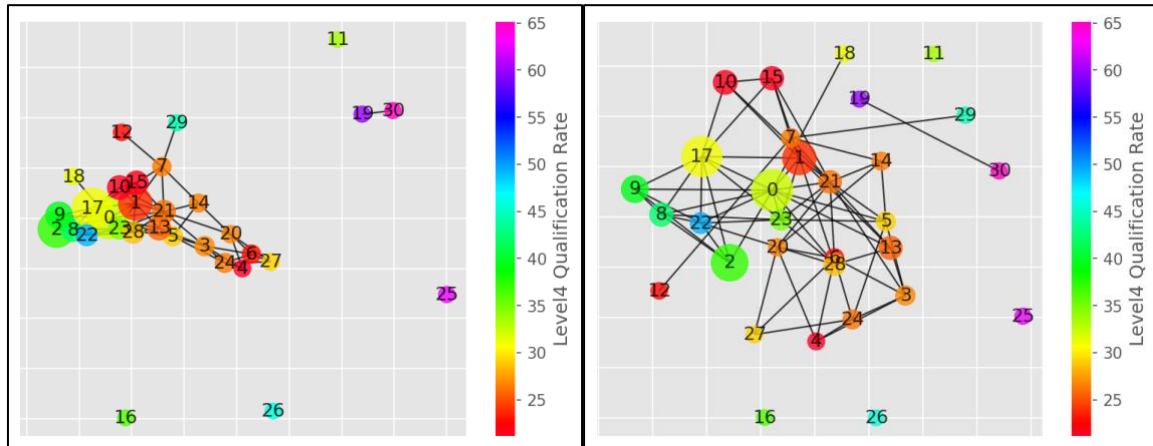


Figure 17. $\epsilon = 15, 20, 25, 30$ Coordination

Figure 18. $k = 0.3, 0.9$ Coordination

To maximize the interpretability of the ball mapper, visual effects were adjusted through the hyperparameters epsilon and k , which represent the distance between nodes. A smaller epsilon value enables more detailed differentiation of balls; however, this may result in a loss of topological connectivity. Conversely, the node length was calibrated to identify the optimal point at which connectivity was most smoothly expressed.

2) Empirical comparative analysis through input variable change

Contrary to conventional assumptions that car ownership, dwelling size, and ethnic composition would correlate with higher levels of educational attainment, the ball mapper results revealed that these factors instead contribute to regional heterogeneity. This empirical divergence from theoretical expectations can be attributed to lifestyle differentiation between urban and rural geographical contexts.

For further analysis, the ball mapper was regenerated using the original set of 28 variables to enhance the statistical robustness of findings by incorporating the entire available dataset.

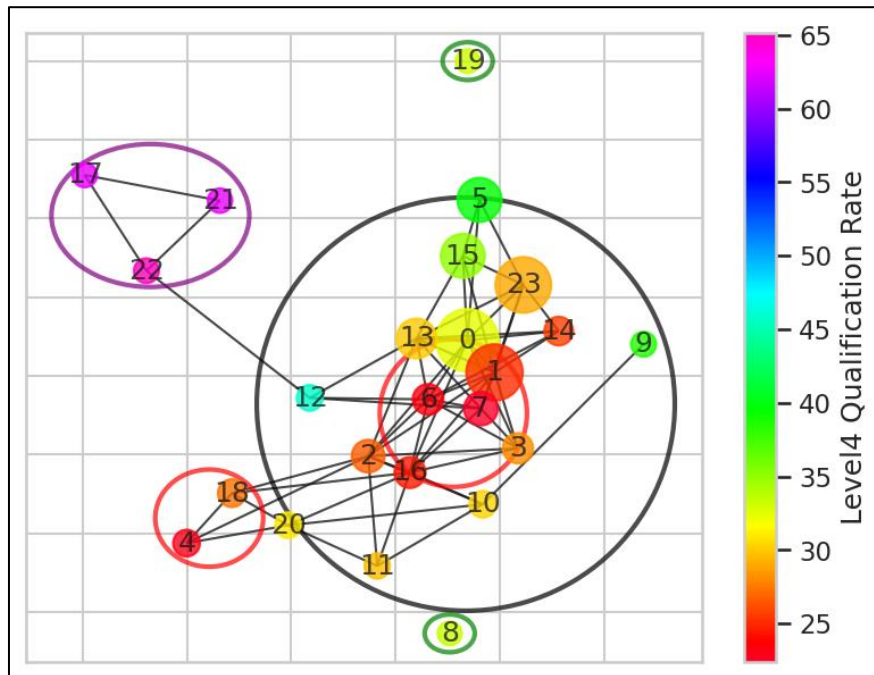


Figure 19. Ball Mapper Level 4 + Qualification rate (28 variables)

The visualization reveals a distinctive radial configuration characterized by a gradient distribution pattern; wherein chromatic intensity progressively amplifies from the central nodes toward the peripheral regions. This spatial arrangement suggests a structure where educational advantages diffuse as distance increases from the characteristics possessed by the centrifugal force.

Central Hub

Balls 1, 6, 7, and 16 are encompassing localities such as Halewood South, Miles Platting & Newton Heath, and Tuebrook and Stoneycroft. These particular nodes exhibit a chromatic gradient indicative of diminished Level 4 Qualification Rates and function as central vertices demonstrating maximum edge connectivity within the network topology. This spatial configuration suggests significant interconnectivity among geographical regions characterized by lower educational attainment indices, potentially indicative of shared socioeconomic determinants.

Lower-left arm

Balls 4 and 18, which encompass the St. Mary's and Longsight areas, exhibit low educational attainment and are spatially distant from the central cluster. This separation is attributed to differences in population density and, significantly, ethnic composition: while the central cluster is predominantly White (average 77.56%), this group is predominantly non-White (average 22.1%), suggesting distinct characteristics of isolation.

Upper-left arm

Balls 17, 21, and 22, encompassing the Deansgate and Blackfriars&Trinity localities, exhibit significantly elevated educational attainment indices and demonstrate substantial interconnectivity. Despite manifesting limited topological connections to the central cluster, these nodes display pronounced internal cohesion, suggesting that these geographical units constitute efficacious subdomains for the identification of elevated educational achievement patterns.

Outliers

Balls 8 (Central) and 19 (Moss Side), though distant and opposite, are far from the center. Central has low homeownership (17.4%), high private rentals (66.9%), and 68.1% unemployment/full-time students. Moss Side is characterized by a high proportion of vulnerable individuals, with a 28.5% deprivation rate (Deprived24p), and a significant 42.9% of social rental housing, leading to its isolation within the network. This may indicate a need for policy intervention.

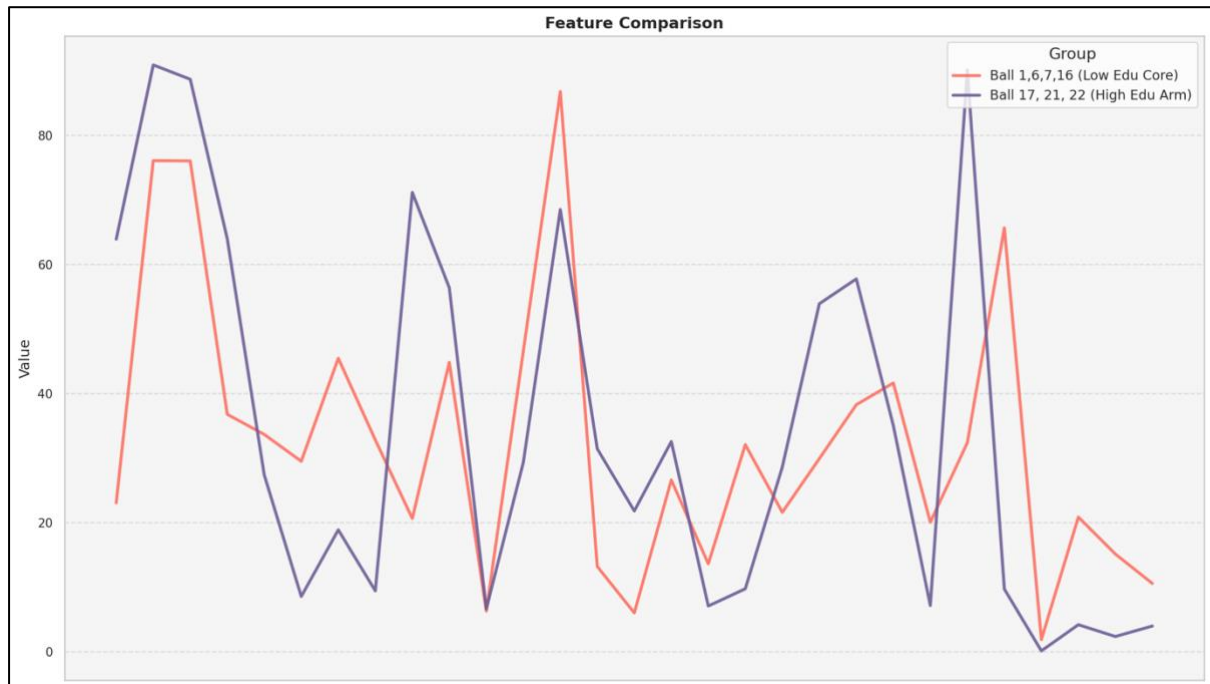


Figure 20. Variables Comparison between Central Low Qualification and High Arm

Variables	Ball 17, 21, 22 (High Edu Arm)	Ball 1,6,7,16 (Low Edu Core)	Difference
high_value	63.9	23.1	40.8
goodhealthp	90.9	76.1	14.8
ndisabledp	88.7	76.1	12.6
Deprived0p	64.0	36.8	27.2
Deprived1p	27.4	33.7	-6.3
Deprived24p	8.6	29.5	-21.0
tenureownp	18.9	45.5	-26.6
socrentp	9.5	32.8	-23.4
privrentp	71.2	20.7	50.5
employeep	56.4	44.8	11.6
selfemployeep	6.7	6.3	0.3
nonemployedp	29.4	46.6	-17.3
ethnwhitep	68.5	86.8	-18.3

ethnotherp	31.5	13.2	18.2
nssec1	21.8	6.0	15.8
nssec2	32.6	26.6	6.0
nssec3	7.1	13.6	-6.5
nssec4	9.8	32.1	-22.3
nssec5	28.6	21.6	7.0
popdensity	53.9	29.9	24.0
nocarp	57.8	38.3	19.5
1carp	35.1	41.7	-6.6
23carp	7.2	20.1	-12.9
lowhousingp	90.1	32.4	57.7
midhousingp	9.7	65.7	-56.0
highhousingp	0.2	1.9	-1.7
youthp	4.2	20.9	-16.7
oldp	2.4	15.2	-12.8
numberofschools	4.0	10.6	-6.6

Table 5. Variables Comparison between Central Low Qualification and High Arm

Several empirical observations contradicted established theoretical expectations. Car ownership indices demonstrated a positive correlation with regions of diminished educational attainment, while deprivation metrics exhibited significantly reduced values in localities characterized by elevated educational attainment. Moreover, 90.1% of high-education clusters resided in residential units comprising 1-2 rooms, whereas low-education nodes displayed a pronounced tendency toward occupation of dwellings with 3-5 rooms. Proprietorship rates were elevated in low-education geographic sectors, with approximately 71.2% of high-education clusters occupying private rental accommodations. These distributional patterns likely reflect urban-rural dichotomization.

Ethnic compositional analysis revealed higher proportional representation of White demographic constituents in low-education regions. Self-reported health status indicators

were superior in high-education areas, though disability prevalence was paradoxically elevated. Additionally, lower educational attainment demonstrated association with higher concentrations of elderly populations (aged 65+) and reduced population density metrics. Notably, the quantitative presence of educational institutions did not exhibit robust correlation with educational achievement indices; indeed, high-achievement localities typically demonstrated reduced educational institutional density.

3) Empirical comparative analysis through output variable change

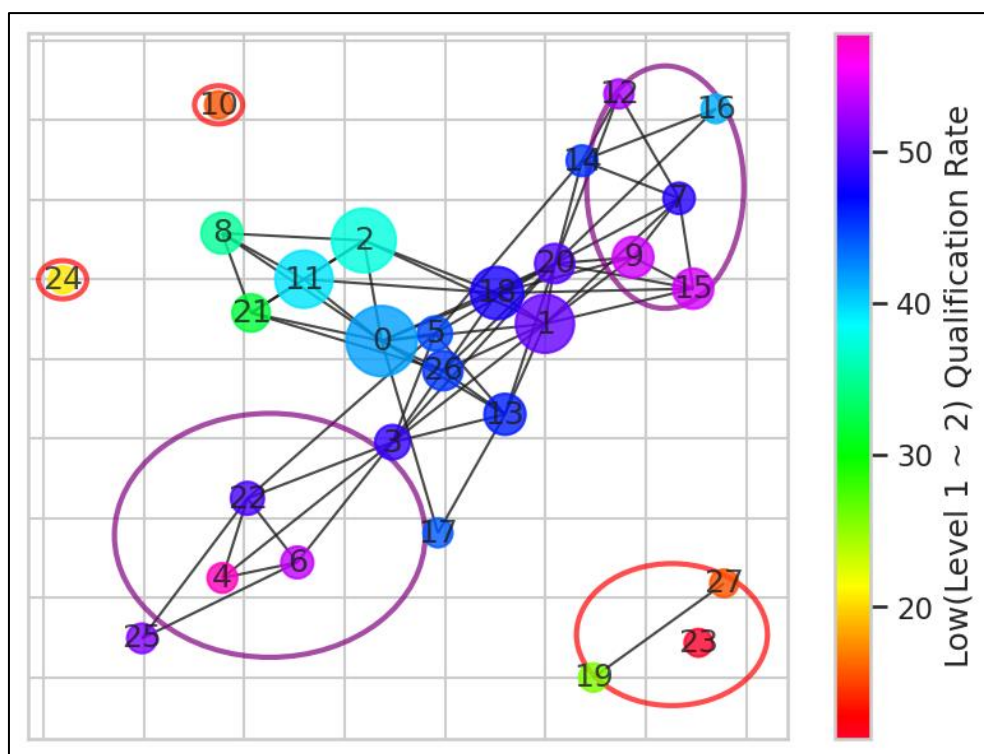


Figure 21. Ball Mapper Level 1~2 Qualification rate (10 variables)

Diagonally upper right quadrant

In the initial interpretative analysis of the ball mapper configuration, the centrally located blue-chromatic nodes, which constitute the predominant connectivity nexus. These represent areas where the education level is low, typically limited to compulsory education. A dominant flow extends diagonally upwards to the right, and at the extreme end of this line are clusters

with even lower qualification rates. Although disconnected from the center, there is also a downward-right diagonal where balls with significantly higher education levels are positioned — they appear as outliers due to their detachment from the main flow. This group reflects a distinct pattern, showing limited connection with the central cluster, highlighting their uniqueness.

The upper-right purple cluster, which includes balls 12, 7, 9, and 15, corresponds to areas such as Everton, Tuebrook, and Stoneycroft. And the lower-left edge of the diagonal — consisting of balls 25, 4, and 6 — includes areas like Longsight, Rusholme, and Alexandra.

The heterogeneity between these groups can be explained by underlying demographic differences. Everton, Tuebrook, and Stoneycroft show high disability rates and a significant aging population, with over 13% aged 65 or older. On the other hand, Longsight, Rusholme, and Alexandra wards have a notably high proportion of non-white ethnic groups, which contributes to their divergence from the others.

Outliers and Cluster with small low education prevalence

The upper-left quadrant, characterized by balls 24 and 10 (encompassing Central and Hulme), exhibits spatial units with a statistically lower prevalence of residents possessing minimal educational credentials. These regions demonstrate marked demographic divergence from the clusters represented by balls 19, 23, and 27 in the lower-right segment—corresponding to Piccadilly, Ancoats & Beswick, and Quays.

Central and Hulme manifest a substantial proportion of economically disengaged or non-employed populace. In contrast, Piccadilly, Ancoats & Beswick, and Quays demonstrate elevated concentrations of individuals occupying managerial professional (up to 64%). This socioeconomic stratification likely reflects endogenous geographic determinants, such as proximal academic institutions or concentrated commercial districts within the urban core.

7. CONCLUSION (Jiaqi)

The intricate interrelationship between socioeconomic determinants and educational attainment indices was manifestly evident, with heterogeneous variables including ethnic demographic composition, health status indicators, geriatric population proportions, and population density metrics demonstrating multidimensional correlations with educational achievement parameters. The finding related to health indicators corresponding Suhrcke et al. (2011)'s analysis, which is, there is a correlation between health status and educational achievements. Addressing educational disparities necessitates a comprehensive, integrated approach, incorporating not solely educational policy frameworks but also social welfare mechanisms, housing initiatives, and broader regional development strategic interventions.

Furthermore, a discernible pattern emerged wherein educational disparities between urban and rural geographical contexts directly contributed to educational inequality gradients, as evidenced by variations in vehicle ownership indices, property proprietorship rates, and residential unit dimensional characteristics. This empirical observation underscores the imperative for targeted policy interventions specifically designed to ameliorate these spatial inequities. These regional inequities substantiate the findings of Van Maarseveen (2021),(2023) Farquharson et al. (2022) and Playford et al. (2023). These empirical results underscore the imperative for policy interventions specifically designed to mitigate these spatial disparities. Such interventions could encompass more equitable resource distribution to rural educational institutions, enhancement of transportation infrastructure, and the expansion of access to digital learning platforms for students situated in geographically isolated areas.

Notably, a direct quantitative correlation between educational institutional density and educational achievement outcomes was not empirically substantiated. This finding potentially corroborates the theoretical position that qualitative enhancement of educational provision should be prioritized over quantitative expansion of educational establishments. Furthermore, the analysis of outliers in Central and Hulme suggests that the nature of the institutions may have a more significant impact on educational outcomes than their sheer number.

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APPENDIX (Jahanvi)

MethodologyProcess of data transformation

Scaling is a very common technique used in machine learning to normalize all the variables allowing for equal representation in the model. However, the dataset that we had included all variables in number and percentage format. Therefore, the first step was to remove all the columns with numbers and keep only percentage columns. Some columns were purely numerical, like number of schools and population density, which were standard scaled [0,1].

Each category that we were working with had more than 5 variables each resulting in a dataset of more than 300 columns. It was not possible to work with such a huge dataset. Moreover, a lot of the variable were redundant. Therefore, we tried to combine as many columns as possible. All the columns concatenated are mentioned below by their categories.

Category 1: Qualification

Original columns:

Qualification_No_Number,	Qualification_No_Percentage,	Qualification_Level_1_Number,
Qualification_Level_1_Percentage,		Qualification_Level_2_Number,
Qualification_Level_2_Percentage,		Qualification_Apprenticeship_Number,
Qualification_Apprenticeship_Percentage,		Qualification_Level_3_Number,
Qualification_Level_3_Percentage,	Qualification_Level_4_Or_Above_Number,	
Qualification_Level_4_Or_Above_Percentage,		Qualification_Other_Number,
Qualification_Other_Percentage		

Final columns: High_value, Below_high_value, Low_value, ratio


```

high = results['Qualification_Level_4_Or_Above_Percentage']
below_high = results[['Qualification_No_Percentage', 'Qualification_Level_1_Percentage',
                    'Qualification_Level_2_Percentage', 'Qualification_Apprenticeship_Percentage',
                    'Qualification_Level_3_Percentage']].sum(axis=1)
low = results[['Qualification_No_Percentage', 'Qualification_Level_1_Percentage',
                'Qualification_Level_2_Percentage']].sum(axis=1)
ratio = high / below_high

results['high_value'] = high
results['below_high_value'] = below_high
results['low_value'] = low
results['ratio'] = ratio

```

Category 2 : Health

Original columns:

Health_Very_Good_Number, Health_Very_Good_Percentage, Health_Good_Number,
 Health_Good_Percentage, Health_Fair_Number, Health_Fair_Percentage,
 Health_Bad_Number, Health_Bad_Percentage, Health_Very_Bad_Number,
 Health_Very_Bad_Percentage

Final columns: goodhealthp

```

results['goodhealthp'] = results[['Health_Very_Good_Percentage', 'Health_Good_Percentage']].sum(axis=1)
results['goodhealthp'].head(3)

```

Category 3: Disability

Original columns:

Disability_Equality_Act_Number Disability_Equality_Act_Percentage
 Disability_Limited_A_Lot_Number Disability_Limited_A_Lot_Percentage
 Disability_Limited_A_Little_Number Disability_Limited_A_Little_Percentage
 Not_Disabled_Equality_Act_Number Not_Disabled_Equality_Act_Percentage
 Not_Disabled_Long_Term_Condition_Number
 Not_Disabled_Long_Term_Condition_Percentage
 Not_Disabled_No_Condition_Number Not_Disabled_No_Condition_Percentage

Final columns: disabledp

```
results['ndisabledp'] = results['Not_Disabled_Equality_Act_Percentage']
results['ndisabledp'].head(1)
```

Category 4: Deprivation

Original columns:

Household_Not_Deprived_Number	Household_Not_Deprived_Percentage
Household_Deprived_One_Dimension_Number	
Household_Deprived_One_Dimension_Percentage	
Household_Deprived_Two_Dimensions_Number	
Household_Deprived_Two_Dimensions_Percentage	
Household_Deprived_Three_Dimensions_Number	
Household_Deprived_Three_Dimensions_Percentage	
Household_Deprived_Four_Dimensions_Number	
Household_Deprived_Four_Dimensions_Percentage	

Final columns: Deprived0p, Deprived1p, Deprived24p

```
results['Deprived0p'] = results['Household_Not_Deprived_Percentage']
results['Deprived1p'] = results['Household_Deprived_One_Dimension_Percentage']
results['Deprived24p'] = results[['Household_Deprived_Two_Dimensions_Percentage', 'Household_Deprived_Three_Dimensions_Percentage',
                                   'Household_Deprived_Four_Dimensions_Percentage']].sum(axis=1)
print(results[['Deprived0p', 'Deprived1p', 'Deprived24p']].mean())
```

Category 5: Tenure

Original columns:

Total_Households_Number	Total_Households_Percentage	Owned_Number
Owned_Percentage	Owned_Outright_Number	Owned_Outright_Percentage
Owned_Mortgage_Number		Owned_Mortgage_Percentage
Shared_Ownership_Number	Shared_Ownership_Percentage	
Shared_Ownership_Detail_Number	Shared_Ownership_Detail_Percentage	
Social_Rented_Number	Social_Rented_Percentage	
Social_Rented_Council_Number	Social_Rented_Council_Percentage	

Social_Rented_Other_Number Social_Rented_Other_Percentage
 Private_Rented_Number Private_Rented_Percentage
 Private_Rented_Landlord_Number Private_Rented_Landlord_Percentage
 Private_Rented_Other_Number Private_Rented_Other_Percentage
 Lives_Rent_Free_Number Lives_Rent_Free_Percentage
 Owns_Mortgage_Shared_Ownership_Number
 Owns_Mortgage_Shared_Ownership_Percentage
 Private_Rented_Lives_Free_Number Private_Rented_Lives_Free_Percentage

Final columns: tenureownp, socrentp, privrentp

```
tenure_columns = results.filter(regex='Own|Rent', axis=1).filter(like='Percentage', axis=1)
tenure_columns.columns
```

```
results['tenureownp'] = results['Owned_Percentage']
results['socrentp'] = results['Social_Rented_Percentage']
results['privrentp'] = results['Private_Rented_Percentage']
```

Category 6: Economic Activity

Original columns:

Economically_Active_Number Economically_Active_Percentage
 In_Employment_Number In_Employment_Percentage
 Employee_Number Employee_Percentage Employee_Part_Time_Number
 Employee_Part_Time_Percentage Employee_Full_Time_Number
 Employee_Full_Time_Percentage Self_Employed_With_Employees_Number
 Self_Employed_With_Employees_Percentage
 Self_Employed_With_Employees_Part_Time_Number
 Self_Employed_With_Employees_Part_Time_Percentage
 Self_Employed_With_Employees_Full_Time_Number
 Self_Employed_With_Employees_Full_Time_Percentage
 Self_Employed_Without_Employees_Number
 Self_Employed_Without_Employees_Percentage

Self_Employed_Without_Employees_Part_Time_Number
Self_Employed_Without_Employees_Part_Time_Percentage
Self_Employed_Without_Employees_Full_Time_Number
Self_Employed_Without_Employees_Full_Time_Percentage
Unemployed_Number Unemployed_Percentage
Economically_Active_Full_Time_Student_Number
Economically_Active_Full_Time_Student_Percentage
Full_Time_Student_Employed_Number Full_Time_Student_Employed_Percentage
Full_Time_Student_Employee_Number Full_Time_Student_Employee_Percentage
Full_Time_Student_Employee_Part_Time_Number
Full_Time_Student_Employee_Part_Time_Percentage
Full_Time_Student_Employee_Full_Time_Number
Full_Time_Student_Employee_Full_Time_Percentage
Full_Time_Student_Self_Employed_With_Employees_Number
Full_Time_Student_Self_Employed_With_Employees_Percentage
Full_Time_Student_Self_Employed_With_Employees_Part_Time_Number
Full_Time_Student_Self_Employed_With_Employees_Part_Time_Percentage
Full_Time_Student_Self_Employed_With_Employees_Full_Time_Number
Full_Time_Student_Self_Employed_With_Employees_Full_Time_Percentage
Full_Time_Student_Self_Employed_Without_Employees_Number
Full_Time_Student_Self_Employed_Without_Employees_Percentage
Full_Time_Student_Self_Employed_Without_Employees_Part_Time_Number
Full_Time_Student_Self_Employed_Without_Employees_Part_Time_Percentage
Full_Time_Student_Self_Employed_Without_Employees_Full_Time_Number
Full_Time_Student_Self_Employed_Without_Employees_Full_Time_Percentage
Full_Time_Student_Unemployed_Number
Full_Time_Student_Unemployed_Percentage Economically_Inactive_Number
Economically_Inactive_Percentage Economically_Inactive_Retired_Number
Economically_Inactive_Retired_Percentage
Economically_Inactive_Student_Number
Economically_Inactive_Student_Percentage

Economically_Inactive_Looking_After_Home_Number
 Economically_Inactive_Looking_After_Home_Percentage
 Economically_Inactive_Long_Term_Sick_Number
 Economically_Inactive_Long_Term_Sick_Percentage
 Economically_Inactive_Other_Number Economically_Inactive_Other_Percentage

Final columns: employeep, selfemployeep, nonemployedp

```
results['employeep'] = results['Employee_Percentage']
results['selfemployeep'] = results[['Self_Employed_With_Employees_Percentage', 'Self_Employed_Without_Employees_Percentage']].sum(axis=1)
results['nonemployedp'] = results[['Unemployed_Percentage', 'Economically_Inactive_Percentage']].sum(axis=1)
results[['employeep', 'selfemployeep', 'nonemployedp']].describe()
```

Category 7: Ethnicity

Original columns:

Asian_Total_Number Asian_Total_Percentage Asian_Bangladeshi_Number
 Asian_Bangladeshi_Percentage Asian_Chinese_Number Asian_Chinese_Percentage
 Asian_Indian_Number Asian_Indian_Percentage
 Asian_Pakistani_Number Asian_Pakistani_Percentage
 Asian_Other_Number Asian_Other_Percentage Black_Total_Number
 Black_Total_Percentage Black_African_Number
 Black_African_Percentage Black_Caribbean_Number
 Black_Caribbean_Percentage Black_Other_Number Black_Other_Percentage
 Mixed_Total_Number Mixed_Total_Percentage Mixed_White_Asian_Number
 Mixed_White_Asian_Percentage Mixed_White_Black_African_Number
 Mixed_White_Black_African_Percentage
 Mixed_White_Black_Caribbean_Number
 Mixed_White_Black_Caribbean_Percentage Mixed_Other_Number
 Mixed_Other_Percentage White_Total_Number White_Total_Percentage
 White_British_Number White_British_Percentage White_Irish_Number
 White_Irish_Percentage White_Gypsy_Number White_Gypsy_Percentage
 White_Roma_Number White_Roma_Percentage White_Other_Number

White_Other_Percentage Other_Total_Number Other_Total_Percentage
 Other_Arab_Number Other_Arab_Percentage Other_Any_Number
 Other_Any_Percentage

Final columns: ethnwhitep, ethnotherp

```
results['ethnwhitep'] = results['White_Total_Percentage']
results['ethnotherp'] = results[['Asian_Total_Percentage', 'Black_Total_Percentage', 'Mixed_Total_Percentage', 'Other_Total_Percentage']].sum(axis=1)
```

Category 8: Socio-economic classification

Original columns:

Higher_Management_Number Higher_Management_Percentage
 Lower_Management_Number Lower_Management_Percentage
 Intermediate_Occupations_Number Intermediate_Occupations_Percentage
 Small_Employers_Number Small_Employers_Percentage
 Lower_Supervisory_Number Lower_Supervisory_Percentage
 Semi_Routine_Number Semi_Routine_Percentage Routine_Number
 Routine_Percentage Never_Worked_Unemployed_Number
 Never_Worked_Unemployed_Percentage
 Full_Time_Students_Number
 Full_Time_Students_Percentage
 Total_Employment_Number
 Total_Employment_Percentage
 Managers_Directors_Number
 Managers_Directors_Percentage
 Professional_Occupations_Number
 Professional_Occupations_Percentage
 Associate_Prof_Tech_Number
 Associate_Prof_Tech_Percentage
 Admin_Secretarial_Number
 Admin_Secretarial_Percentage
 Skilled_Trades_Number
 Skilled_Trades_Percentage
 Caring_Leisure_Number
 Caring_Leisure_Percentage
 Sales_Customer_Service_Number
 Sales_Customer_Service_Percentage
 Process_Plant_Number
 Process_Plant_Percentage
 Elementary_Occupations_Number
 Elementary_Occupations_Percentage

Final columns: nssec1, nssec2, nssec3, nssec4, nssec5

```

results[['Higher_Management_Percentage',
        'Lower_Management_Percentage', 'Intermediate_Occupations_Percentage',
        'Small_Employers_Percentage', 'Lower_Supervisory_Percentage',
        'Semi_Routine_Percentage', 'Routine_Percentage',
        'Never_Worked_Unemployed_Percentage', 'Full_Time_Students_Percentage']]

results['nssec1'] = results[['Higher_Management_Percentage']]
results['nssec2'] = results[['Lower_Management_Percentage', 'Intermediate_Occupations_Percentage']].sum(axis=1)
results['nssec3'] = results[['Small_Employers_Percentage', 'Lower_Supervisory_Percentage']].sum(axis=1)
results['nssec4'] = results[['Semi_Routine_Percentage', 'Routine_Percentage']].sum(axis=1)
results['nssec5'] = results[['Never_Worked_Unemployed_Percentage', 'Full_Time_Students_Percentage']].sum(axis=1)

```

Category 9: Housing

Original columns:

totalallhouseholds.x, percentallhouseholds.x, num1bedroom, percent1bedroom,
 num2bedrooms, percent2bedrooms, num3bedrooms, percent3bedrooms, num4bedrooms,
 percent4ormorebedrooms, num1room, percent1room, num2rooms, percent2rooms
 num3rooms, percent3rooms, num4rooms, percent4rooms, num5rooms,
 percent5rooms num6rooms percent6rooms, num7rooms, percent7rooms, num8rooms,
 percent8rooms, num9ormorererooms, percent9ormorererooms, occupancy2ormorebedrooms,
 percent2ormorebedrooms, occupancy1bedroom, percentoccupancy1bedroom,
 occupancy0bedrooms, percentoccupancy0bedrooms, occupancy1.1bedrooms,
 percentoccupancy1.1bedrooms, occupancy2orlessbedrooms, percent2orlessbedrooms,
 totalallhouseholds.y.y, percentallhouseholds.y.y, occupancy2ormorererooms,
 percent2ormorererooms, occupancy1room, percentoccupancy1room, occupancy0rooms,
 percentoccupancy0rooms, occupancy1.1rooms, percentoccupancy1.1rooms,
 occupancy2orlessrooms, percent2orlessrooms

Final columns: lowhousingp, midhousingp, highhousingp

```

results['lowhousingp'] = results[['percent1room', 'percent2rooms', 'percent3rooms']].sum(axis=1)
results['midhousingp'] = results[['percent4rooms', 'percent5rooms', 'percent6rooms']].sum(axis=1)
results['highhousingp'] = results[['percent7rooms', 'percent8rooms', 'percent9ormorererooms']].sum(axis=1)

results[['lowhousingp', 'midhousingp', 'highhousingp']].describe()

```

Category 10: Car Ownership

Original columns:

Cartotn, cartotp, nocar, nocarp, X1car, X1carp, X2car, X2carp, X3car, X3carp

Final columns: nocarp, 1carp, 23carp

```
results['nocarp']
results['1carp'] = results['X1carp']
results['23carp'] = results[['X2carp', 'X3carp']].sum(axis=1)
```

Category 11: Age

Original columns:

Aged.4.years.and.under, Aged.5.to.9.years, Aged.10.to.15.years, Aged.16.to.19.years,
Aged.20.to.24.years, Aged.25.to.34.years, Aged.35.to.49.years, Aged.50.to.64.years,
Aged.65.to.74.years, Aged.75.to.84.years, Aged.85.years.and.over

Final columns: youthp, adultp, oldp

```
results['youth'] = results[['Aged.4.years.and.under', 'Aged.5.to.9.years', 'Aged.10.to.15.years']].sum(axis=1)
results['adult'] = results[['Aged.16.to.19.years', 'Aged.20.to.24.years', 'Aged.25.to.34.years', 'Aged.35.to.49.years', 'Aged.50.to.64.years']].sum(axis=1)
results['old'] = results[['Aged.65.to.74.years', 'Aged.75.to.84.years', 'Aged.85.years.and.over']].sum(axis=1)
```

```
results['total_population'] = results[['youth', 'adult', 'old']].sum(axis=1)

results['youth_ratio'] = results['youth'] / results['total_population']
results['adult_ratio'] = results['adult'] / results['total_population']
results['old_ratio'] = results['old'] / results['total_population']

results['youthp'] = results['youth_ratio'] * 100
results['adultp'] = results['adult_ratio'] * 100
results['oldp'] = results['old_ratio'] * 100

results[['youthp', 'adultp', 'oldp']].describe()
```

Category 12: Schools

Original columns: numberofschools

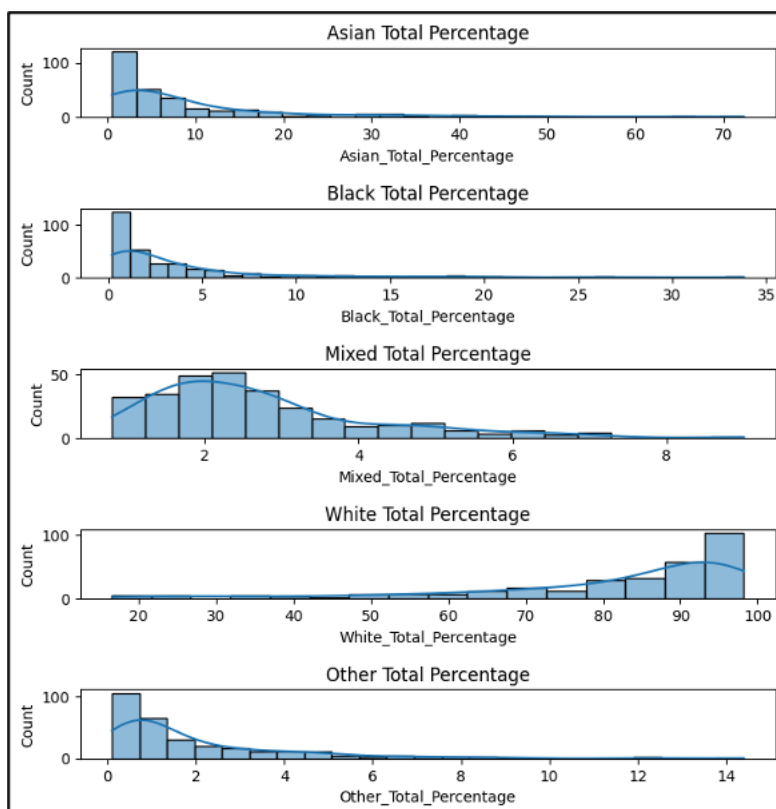
Final columns: numberofschools

A few columns with insufficient data were removed as well. One example is open spaces

(Allotments Or Community Growing Spaces, Bowling Green, Cemetery, Golf Course, Other Sports Facility, Play Space, Playing Field, Public Park Or Garden, Religious Grounds, Tennis Court).

Rationale for choosing 10 variables

Out of the 28 variables, we only worked with 10 to ensure relevancy of results (i.e., each category was represented by one variable). This rationale can be explained by using an example, take ethnicity. After exploring the columns within this variable, we see that Asian, Black, Mixed, and Other show a very similar trend with a peak at the left side of the graph, closer to 0. However, White shows a completely different trend with its peak at the opposite end of the graph.



This then motivated us to create a variable called `ethnotherp` which added the values from Asian,

Black, Mixed, and Other ethnicities (see attached code snippet).

```
results['ethnwhite'] = results['White_Total_Percentage']
results['ethnotherp'] = results[['Asian_Total_Percentage', 'Black_Total_Percentage', 'Mixed_Total_Percentage', 'Other_Total_Percentage']].sum(axis=1)
```

A similar approach was taken with nonemployedp where it refers to the percentage of unemployed individuals and economically inactive individuals.

```
results['employeeep'] = results['Employee_Percentage']
results['selfemployeeep'] = results[['Self_Employed_With_Employees_Percentage', 'Self_Employed_Without_Employees_Percentage']].sum(axis=1)
results['nonemployedp'] = results[['Unemployed_Percentage', 'Economically_Inactive_Percentage']].sum(axis=1)
results[['employeeep', 'selfemployeeep', 'nonemployedp']].describe()
```

Midhousingp, socrentp, and oldp all followed the same logic.

Extra Visualisations

